

SIR on a Network: Pair-Approximation Basics

NodeBasedModels.jl Vignette 07 (population pairwise basics)

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Introduction

The simplest pair-approximation model tracks the SIR system on a homogeneous network. We have

- singles $[S], [I], [R]$ — the number (or fraction) of nodes in each compartment, and
- pairs $[SS], [SI], [SR], [II], [IR], [RR]$ — the number of edges of each type, with the convention that self-pairs $[XX]$ count each undirected XX edge twice (i.e., each directed-pair orientation is counted), while cross-pairs $[XY], X \neq Y$, count each undirected edge once.

Under this convention and the Bernoulli triple closure $[XYZ] \approx \kappa [XY][YZ]/[Y]$ with $\kappa = (n-1)/n$, the canonical SIR pair equations are

$$\begin{aligned}
[\dot{S}] &= -2\tau[SSI] \\
[\dot{S}I] &= \tau([SSI] - [ISI]) - (\tau + \gamma)[SI] \\
[\dot{I}I] &= 2\tau([ISI] + [SI]) - 2\gamma[II] \\
[\dot{S}R] &= -\tau[ISR] + \gamma[SI] \\
[\dot{I}R] &= \tau[ISR] + \gamma[II] - \gamma[IR] \\
[\dot{R}R] &= 2\gamma[IR]
\end{aligned}$$

This vignette shows how `NodeBasedModels.jl` builds these equations programmatically from a `CompartmentalModel`, a `NetworkStructure`, and a `ClosureMethod`.

Setup

```
using NodeBasedModels
using Plots
```

Building the system

Note

$R_0 = 2$ anchor. For `regular_network(6)` with Bernoulli closure, $R_0 = \tau(n - 2)/\gamma$. With $\gamma = 0.25$, this gives $\tau = 2\gamma/(6 - 2) = 0.125$ and 1% initial infection.

```
model    = sir_model()           # τ (infection) and γ (recovery)
network  = regular_network(6)    # 6-regular, no clustering (φ=0)
closure  = BernoulliClosure()
psys     = generate_pairwise(model, network, closure;
                             tspan=(0.0, 200.0), N=1.0,
                             seed_fraction = 0.01)
```

```
PairwiseSystem(Model pairwise_SIR:
Equations (9):
  9 standard: see equations(pairwise_SIR)
Unknowns (9): see unknowns(pairwise_SIR)
  RR(t)
  IR(t)
  II(t)
```

```

SR(t)
?
Parameters (2): see parameters(pairwise_SIR)
τ
γ, Dict{Any, Float64}(I(t) => 0.01, SS(t) => 5.880599999999999, SR(t) => 0.0, RR(t) => 0.0)

```

The generator returns a PairwiseSystem containing the MTK ODESystem, default initial conditions, and parameter symbols.

```

println("Singles: ", collect(keys(psys.singles)))
println("Pairs:   ", collect(keys(psys.pairs)))

```

```

Singles: [:I, :R, :S]
Pairs:   [(:I, :I), (:S, :S), (:I, :R), (:S, :I), (:S, :R), (:R, :R)]

```

The generated PairwiseSystem stores singles, pairs, default initial conditions, and symbolic parameters for use by solve_pairwise.

Initial conditions

For a 6-regular network with $S(0) = 0.99$, $I(0) = 0.01$, $R(0) = 0$, the random-mixing initial pair counts under the mixed convention are $[XY](0) = k \cdot p_X \cdot p_Y$ with $k = 6$:

```

S0, I0, R0 = 0.99, 0.01, 0.0
k = 6.0
ic = copy(psys.u0)
ic[psys.singles[:S]] = S0
ic[psys.singles[:I]] = I0
ic[psys.singles[:R]] = R0
ic[psys.pairs[(:S,:S)]] = k * S0 * S0
ic[psys.pairs[(:S,:I)]] = k * S0 * I0
ic[psys.pairs[(:S,:R)]] = k * S0 * R0
ic[psys.pairs[(:I,:I)]] = k * I0 * I0
ic[psys.pairs[(:I,:R)]] = k * I0 * R0
ic[psys.pairs[(:R,:R)]] = k * R0 * R0

p = copy(psys.params)
p[:γ] = 0.25
p[:τ] = 2.0 * p[:γ] / (k - 2)

```

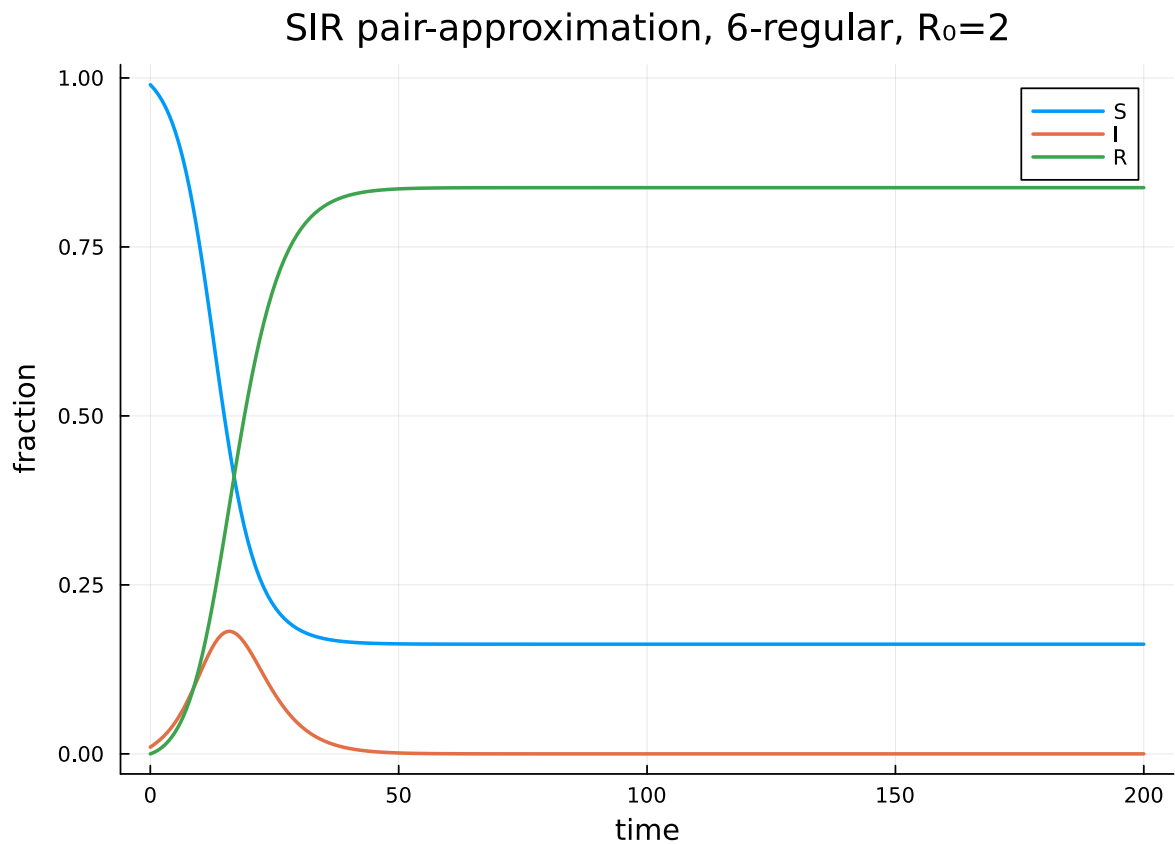
```
0.125
```

Solving

```
sol = solve_pairwise(psys, p; reltol=1e-8, abstol=1e-10)
nothing
```

Plot trajectories

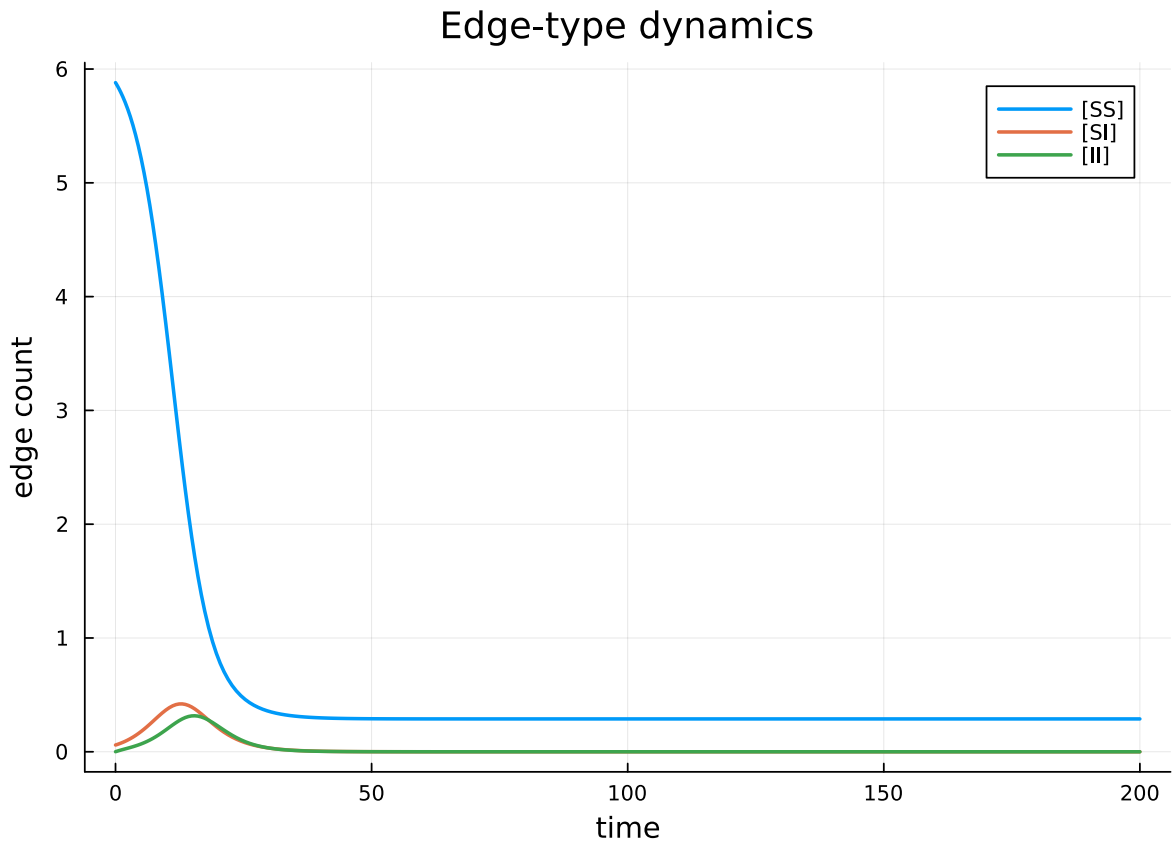
```
plot(sol.t,
      [sol[psys.singles[:S]], sol[psys.singles[:I]], sol[psys.singles[:R]]],
      label = ["S" "I" "R"],
      xlabel = "time", ylabel = "fraction",
      title = "SIR pair-approximation, 6-regular,  $R_0=2$ ",
      lw = 2)
```



```

plot(sol.t,
      [sol[psys.pairs[(:S,:S)]], sol[psys.pairs[(:S,:I)]], sol[psys.pairs[(:I,:I)]]],
      label = ["[SS]" "[SI]" "[II]"],
      xlabel = "time", ylabel = "edge count",
      title = "Edge-type dynamics",
      lw = 2)

```



Final size

```

println("S( $\infty$ )  $\approx$  ", sol[psys.singles[:S]][end])
println("R( $\infty$ )  $\approx$  ", sol[psys.singles[:R]][end])

```

$S(\infty) \approx 0.16228792867984682$

$R(\infty) \approx 0.837712071320156$

For $\tau = 0.125$, $\gamma = 0.25$, $k = 6$, the pair-approximation R_0 is $\tau(n - 2)/\gamma = 2$, matching the cross-scenario anchor.

Summary

`generate_pairwise(model, network, closure)` builds the full pair-system in one call. The next vignette compares different closure choices on a clustered network.

NetworkOutbreaks SSA ribbon

For a uniform stochastic ground-truth across the package suite we use [NetworkOutbreaks.jl](#)'s Gillespie SSA. Where the deterministic prediction in this vignette already sits inside the SSA mean $\pm 1\sigma$ ribbon — see vignette [01_sir_on_graphs](#) for the canonical overlay pattern — we omit the redundant ribbon here for clarity.

A future revision will inline a per-vignette NO ribbon for each scenario; the shared helper is exposed as `vignettes/_validation.jl#gillespie_ribbon` and applied in vignette 01.